1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In HER2-positive breast cancer, a high stromal-to-tumor ratio (TSR) establishes an ECM-rich, angiogenic niche that elevates serum VEGF and promotes tumor cell intravasation, reflected by an unfavorable CTC response during HER2-targeted therapy. Consequently, tumors with high TSR, elevated VEGF, and poor CTC clearance will exhibit diminished therapy_response, identifying a subset that may benefit from adjunctive anti-angiogenic or stromal-targeting interventions.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	serum VEGF concentration, CTC response
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

serum VEGF concentration: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

CTC response: CTC response ↔ pCR path (Core Path 2) supplies the dissemination dynamics that determine whether local tumor control translates into systemic benefit.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Breast Cancer Index (BCI)	Supporting modifier from knowledge graph
Secondary	70-gene assay	Supporting modifier from knowledge graph

Secondary	95GC-H	Supporting modifier from knowledge graph
Secondary	Drug-PIN	Supporting modifier from knowledge graph
Secondary	ICG-R15	Supporting modifier from knowledge graph
Secondary	CPS ≥ 5 group	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Clinical correlative study: quantify TSR in diagnostic/post-treatment tumor sections by histology, measure baseline and on-treatment serum VEGF by ELISA, monitor CTC burden/kinetics using established platforms (e.g., CellSearch or microfluidic capture), and correlate with objective therapy_response (pCR or radiologic/biochemical response) in HER2+ cohorts.

Prediction 2: Preclinical validation: employ HER2+ breast cancer models with manipulated stromal content (e.g., co-culture with fibroblasts or ECM-modulating enzymes; or patient-derived xenografts with variable ECM density) to measure VEGF secretion, CTC surrogates, and response to HER2 inhibitors alone and with anti-angiogenic/stroma-targeting agents.

Prediction 3: Mechanistic experiments: assess whether TSR- or VEGF-driven changes in vascular permeability and ECM remodeling mediate CTC extravasation using in vivo imaging (DCE-MRI, IVIS for labeled cells) and ECM remodeling assays.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Breast Cancer Index (BCI): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- 70-gene assay: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.